Interpretation of GLM regression coefficients

Eg: 8.1 $N \sim \text{Poisson}(\mu_i)$ log link , N = no. of stressful events

① $\log(\hat{\mu}_i) = 2.803 - 0.084 x_i$, x_i = month

$\hat{\mu}_i = e^{\hat{\eta}_i}$

$\hat{\beta}_1 = -0.084$, $e^{-0.084} = 0.9196$

⇒ There is 8.04% decrease in the expected no. of stressful events when 1 unit increase in month.

② $\log(\hat{\mu}_i) = 2.803 - 0.084 x_1 + 0.12 x_2$, x_1 = month, x_2 = weight (kg)

$\hat{\beta}_1 = -0.084$, $e^{-0.084} = 0.9196$

⇒ Holding x_2 (weight) constant, there is 8.04% ... ..

Eg. 8.3 logit link

$\log\left(\frac{\hat{\pi}_i}{1-\hat{\pi}_i}\right) = -20.272 + 4.73 x_i$, π_i = probability of death, x_i = dose

$\hat{\beta}_1 = 4.73$, $e^{4.73} = 113.647$

⇒ There is 13.65% increase in the odds of getting death in 1 unit increase in dose

Additional example:

① $\log\left(\frac{\hat{p}_i}{1-\hat{p}_i}\right) = -1.93 + 0.38 I\{\text{smoking}=1\}$, $\hat{p}_i$ = probability of having heart disease in the next 10 years

smoking = {1 , Yes; 0 , No

$\hat{\beta}_1 = 0.38$, $e^{\hat{\beta}_1} = 1.46$

Baseline: non-smoking

⇒ The smoking group has 1.46 times the odds of the non-smoking group of having heart disease.

OR ⇒ There is a 46% greater relative risk of having heart disease in the smoking group compared to the non-smoking group

OR ⇒ The smoking group has 46% more odds of having heart disease than the non-smoking group.

- If $\beta = -0.38$, $e^{-0.36} = 0.68$

  ⇒ There is a 32% reduction in the relative risk of having heart disease in the smoking group compared to the non-smoking group.

- $\hat{\beta}_0 = -1.93$, $e^{-1.93} = 0.13$ ⇒ The probability that a non-smoke will have a heart disease in the next 10 years is 0.13.